AI Capability Boundary Reference

AI Literacy for the Modern Workforce

Know what the tool does well and where it systematically fails before you delegate. The framework calls this Platform Awareness, part of Delegation (Dakan & Feller, 2025, p. 4). It sets two requirements for automation: a clear task definition and a quality check (p. 2). It is organized by task type, not technical architecture. Each entry connects to Module 3's mechanical framework.

WHERE AI IS RELIABLE

Tasks where models produce useful output under well-specified prompts. Reliability here means errors are infrequent rather than absent.

TASK TYPE	WHY IT WORKS	NOTES
Drafting and composition	Dense training patterns for written formats (emails, memos, reports). Well-specified prompts produce well-structured output.	Quality scales with Description quality. Specify product, process, and performance.
Summarizing and compression	The model excels at identifying high-probability patterns, which is what compression requires.	Verify completeness; the model may drop low-frequency details.
Reformatting and restructuring	Tasks with one correct output and a clear specification. Token-level manipulation within known patterns.	Format + audience + length spec → reliable output.
Brainstorming and ideation	Sampling produces varied options; asking for many distinct options widens the spread.	Useful for breadth. Evaluate each option independently; plausible ≠ good.
Translation (major languages)	Dense training representation for high-resource language pairs (English ↔ Spanish, French, German, Mandarin).	Reliability drops for lower-resource languages due to tokenization asymmetry.
Code generation (common patterns)	Well-documented languages and frameworks are heavily represented in training data.	Edge cases, niche libraries, and novel architectures are less reliable. Test all output.
Pattern recognition in text	Categorization, sentiment analysis, entity extraction: tasks where the model identifies structure.	Reliable for well-defined categories. Less reliable for subjective or culturally specific judgments.

WHERE AI SYSTEMATICALLY FAILS

Three categories of unreliable tasks from Module 3. Each traces to a mechanical root cause.

BOUNDARY TASKS *Root cause: Tokenization*

Token boundaries misalign with the meaningful structure of the input. Errors concentrate where tokens and meaning diverge.

FAILS AT...	WHY	INSTEAD...
Multi-digit arithmetic	Digits are cut into tokens by a fixed rule, not by value: "1279" becomes ["127", "9"], so token boundaries cut across place value	**Ask the model to write code that performs the calculation**
Character counting and spelling	The model sees tokens rather than letters; "strawberry" is not processed letter by letter	**Use programmatic tools for character-level tasks**

Non-English text processing	Lower-resource languages often take 2× or more the tokens for the same meaning (up to 15×; Petrov et al., 2023), so they use up context faster	**Verify non-English output more carefully; expect quality gaps**

SPECIFICITY TASKS *Root cause: Next-token prediction × Knowledge*

The model generates statistically probable tokens. When a task requires specific facts the training data represents thinly, it produces plausible fabrications.

FAILS AT...	WHY	INSTEAD...
Specific citations and references	Citation-shaped text is common in training data, so the model generates format-perfect references that may not exist	**Verify every citation against the original source**
Precise statistics and figures	Numbers generated to satisfy the pattern, not retrieved from a database	**Check every specific number against primary data**
Legal / regulatory references	Regulation numbers, case names, and statutory language are specificity-zone content	**Cross-reference against official legal sources**
Named quotes and attributions	Quote-shaped text attributed to specific people is generated by probability, not recall	**Confirm the quote exists in the attributed source**

VOLUME TASKS *Root cause: Context window*

The model has a finite processing window. Information outside is not processed. Information in the middle receives less attention.

FAILS AT...	WHY	INSTEAD...
Long document analysis (>20 pages)	Middle sections get the least attention (the lost-in-the-middle effect): processed but underweighted	**Break long documents into sections; process separately**
Sustained instruction fidelity	Over many turns, earlier instructions lose influence as context fills	**Restate critical instructions periodically; compare to original spec**
Multi-document synthesis	Several documents fill the context fast; those in the middle are most likely to be underprocessed	**Process documents individually, then synthesize yourself**

THE CORE DISTINCTION

The model generates text rather than retrieving facts. Fluent output and accurate output are independent properties. Old software automated anything you could describe in logical steps and gave the same answer every time. This software automates anything with enough training data and anything where verification scales, and it can be wrong (Evans, May 2026, slide 52). Most of the Reliable list is pattern work the training data covers densely. Only the Specificity failures trace to thin training data; the Boundary and Volume failures trace to tokenization and the context window.

Framework: 4D AI Fluency Framework, Delegation (Platform Awareness) and Discernment competencies (Dakan & Feller, 2025, pp. 4–6), as taught in Anthropic's AI Fluency course. Mechanical explanations from Module 3.

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